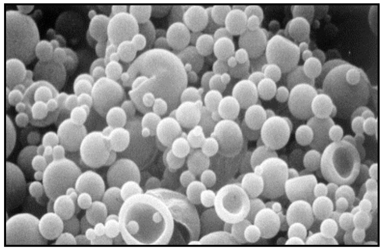
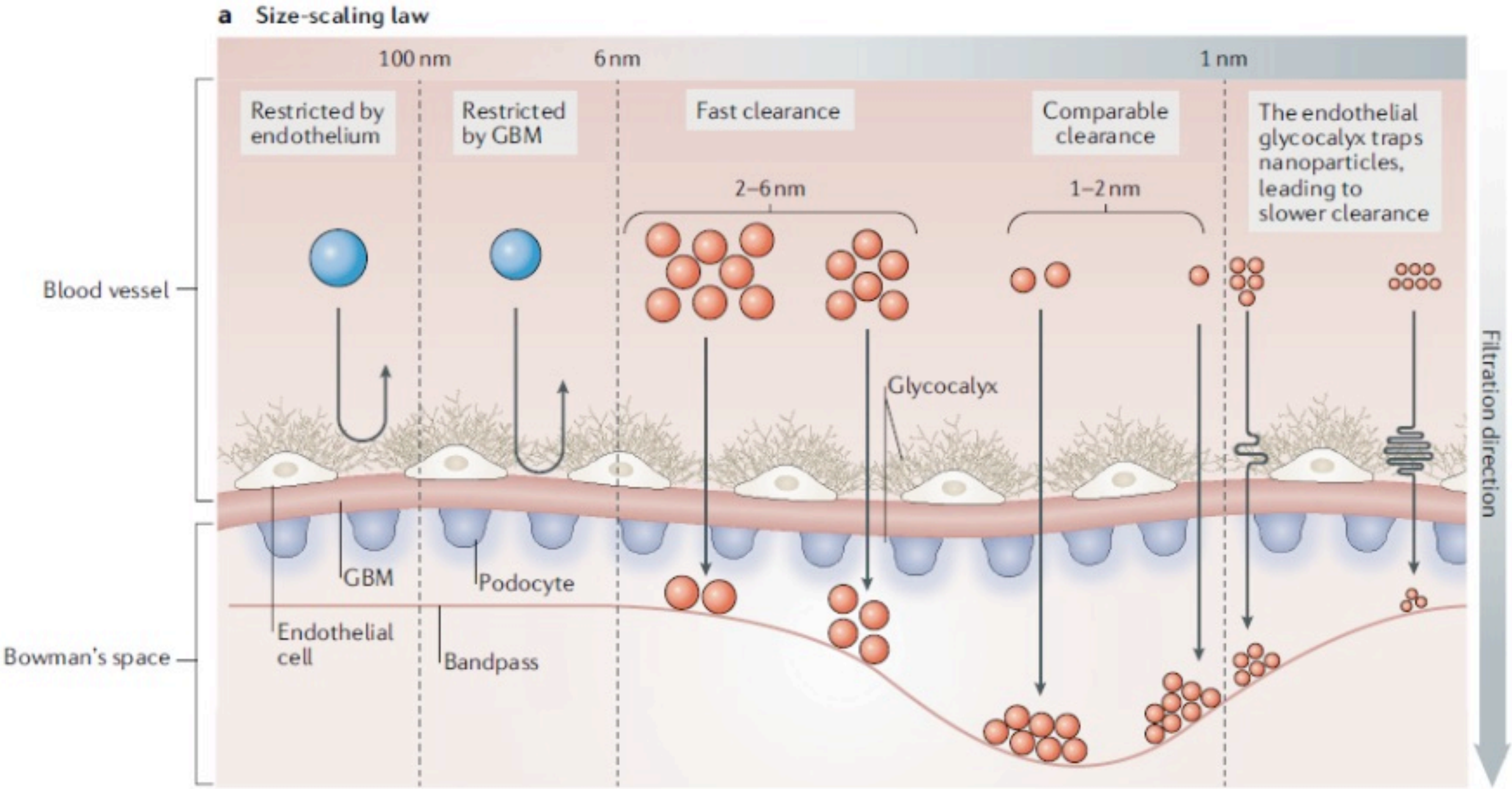
Nano- and Micro-particles

- Acts as a controlled release depot system similar to matrices, reservoirs or hydrogels
- Decreasing the non-specific delivery of the drug to non- target tissues
- Injectable (no need for surgical implantation)
- Convenient way for delivering large hydrophilic molecules inside cells
- Particles can be adapted for targeted delivery to cells and tissues
- Improving the stability of the drug in vivo
- Improving shelf life of the product.

Particles in drug delivery

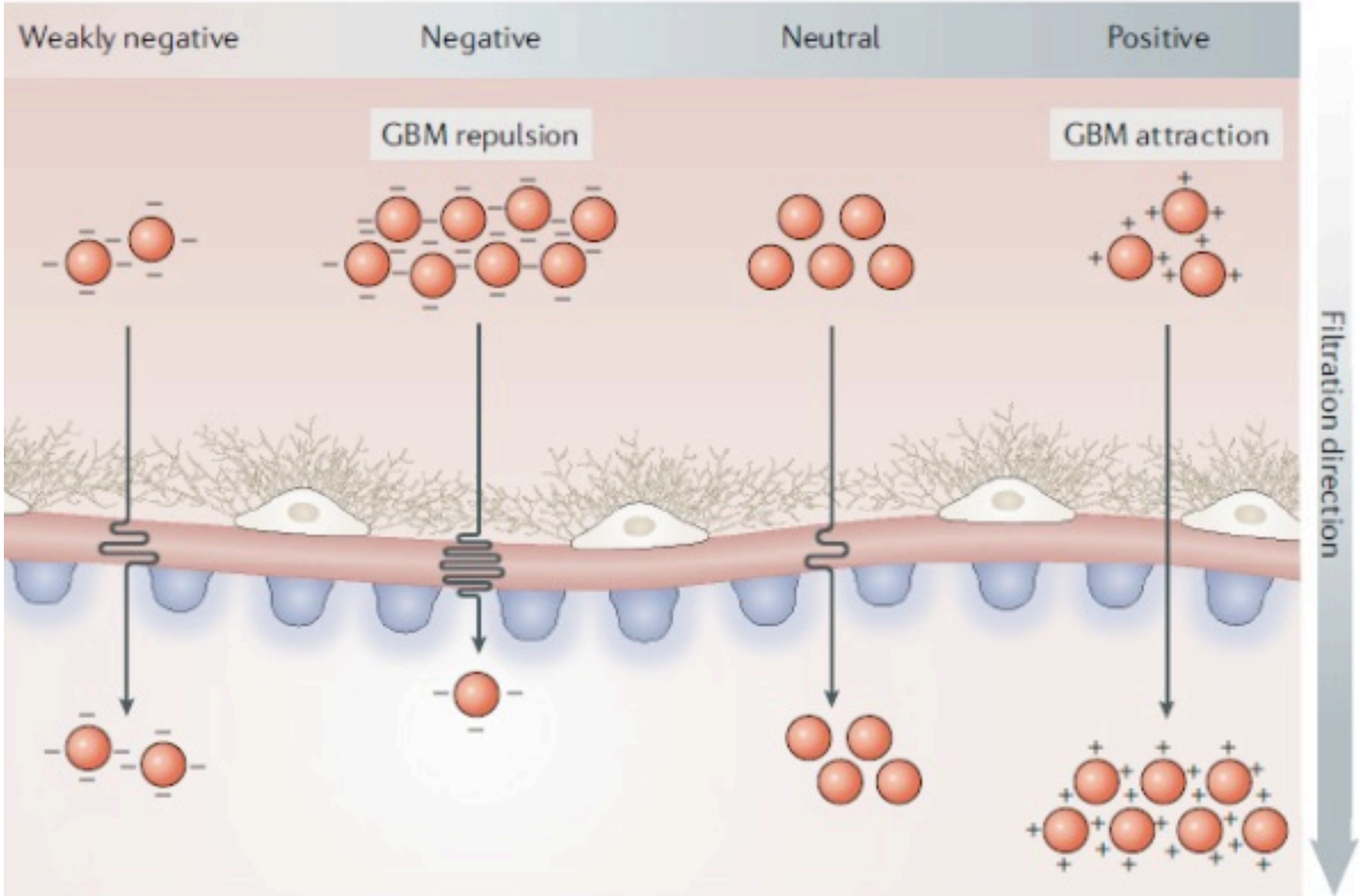


- Microparticles: > 1 μm in size (typically 1-100 μm)
- Nanoparticles: Typically 10nm – 1000nm in size
- Drugs (small molecules, proteins, peptides, nucleic acids) are generally encapsulated within these particles, but they can also be adsorbed or covalently linked to the particle surface
- These particles can be in the form of
  - Micro or nano capsules (well defined polymer envelop surrounding a hollow core)
  - Solid micro or nanoparticles (particles that are essentially a matrix)



- Nanoparticles with a hydrodynamic diameter (HD) > 100 nm cannot traverse the endothelium.
- Nanoparticles with HD = 6–100 nm cannot cross the glomerular basement membrane (GBM).
- Nanoparticles with HD = 1–6 nm can traverse the glomerular filtration membrane, with smaller nanoparticles crossing faster than larger ones.
- Nanoparticles with HD < 1 nm physically interact with the endothelial glycocalyx, resulting in inverse size- dependent filtration, with smaller particles being cleared more slowly than larger ones.

Nature Reviews | **Materials,** https://doi.org/10.1038/s41578-018-0038-3



- The endothelial glycocalyx (charge layer 1), GBM (charge layer 2) and podocyte glycocalyx (charge layer 3) are negatively charged and thus make the glomerular filtration of engineered nanoparticles charge-dependent
- Positively charged nanoparticles are cleared faster than neutral ones, followed by negatively charged nanoparticles, owing mainly to interactions with the GBM

Nature Reviews **| Materials,** https://doi.org/10.1038/s41578-018-0038-3

Particle-mediated delivery: Definitions

- **Particle size and shape**
  - Qualitatively: microscopic techniques. Most commonly by electron microscopic techniques.
  - Quantitative: coulter counting or light scattering techniques. Gives a distribution of diameters from which number average, weight average, volume average sizes can be calculated.

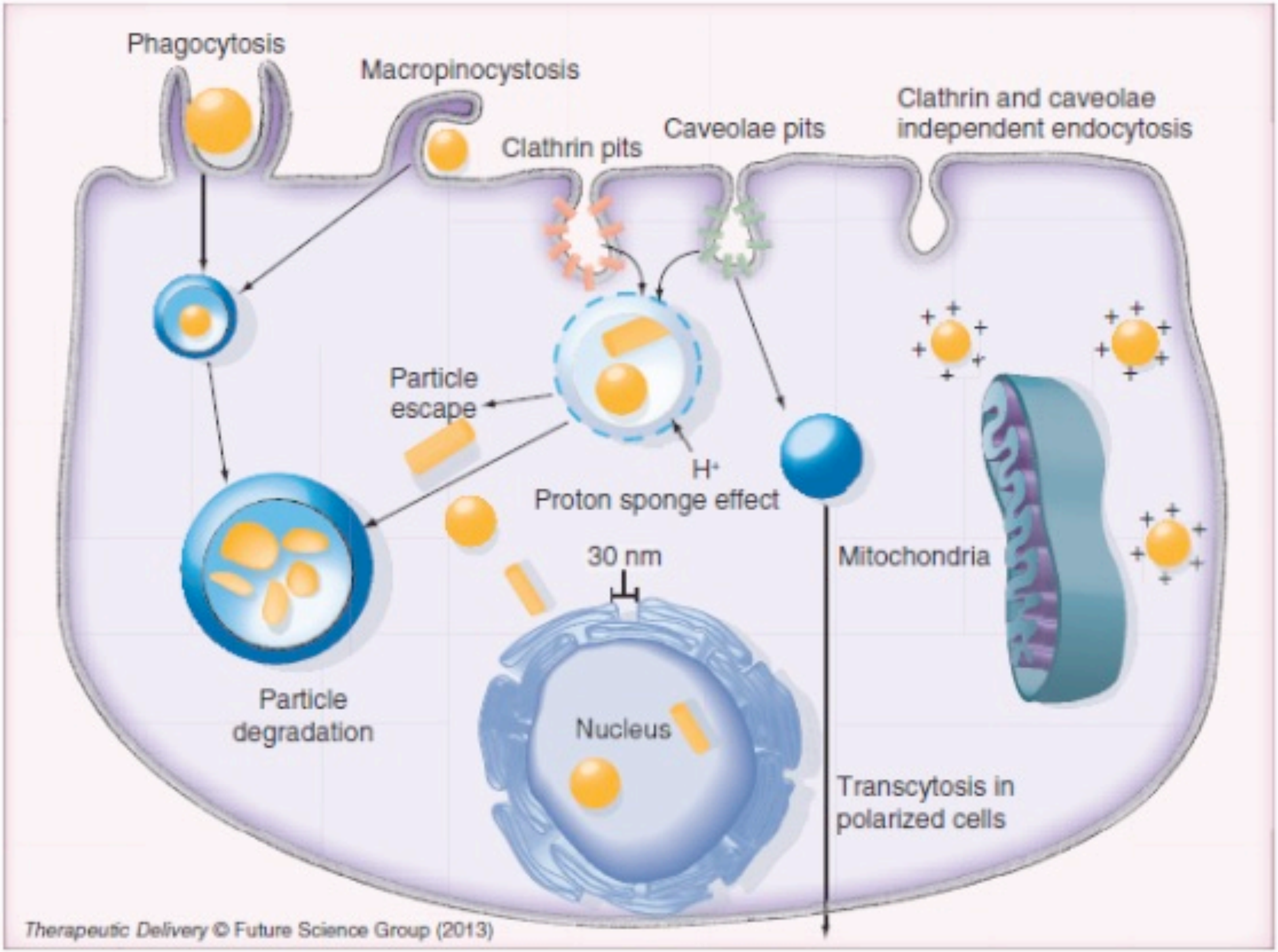
- **Polydispersity**
  - Calculated from the size distribution. Has same definition as polymers. A measure of how broad or narrow the size distribution is.

- **Carrier Composition**
  - Percentage of polymer, drug, solvents, surfactants and any other additives remaining in the final particle formulation

- **Encapsulation efficiency**
  - Percentage of the starting encapsulant (drug) that is captured in the finished polymer particles

- **Loading level or encapsulation ratio**
  - The weight percentage of drug in the particle formulation

Intra-cellular Delivery

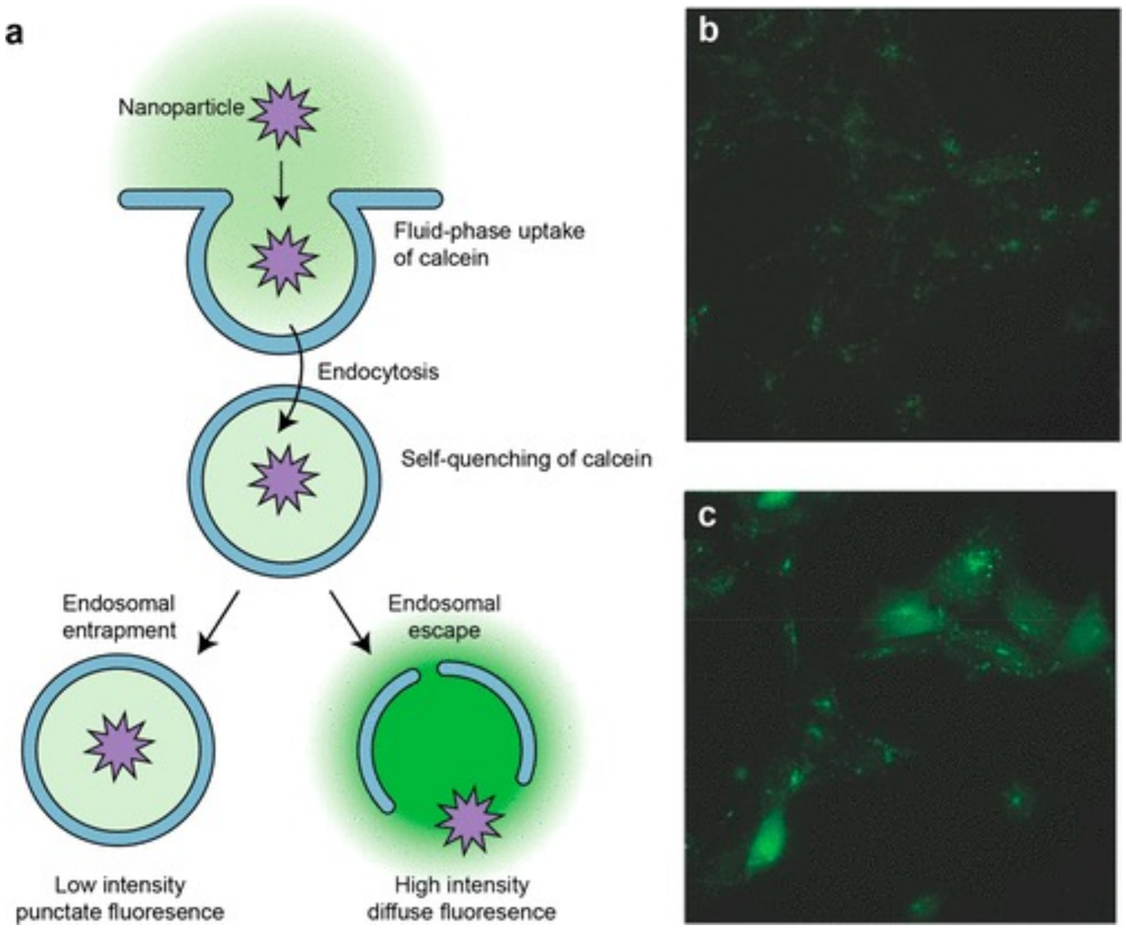


Proton Sponge effect

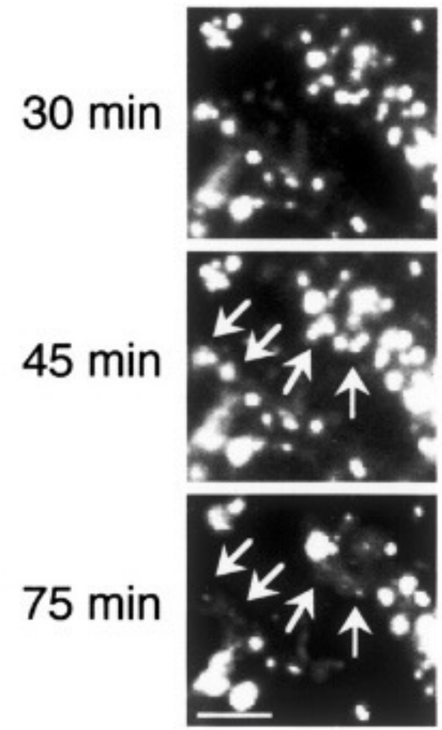
- Particles uptaken through endocytosis
- Get localized in endosomes and lysosomes where the environment is very harsh
- To help escape the particles, “proton sponge” effect is used
- Particles are designed to carry secondary or tertiary amines
  - Buffer the pH of endosomes
  - Results in proton influx through the endosomal proton pumps,
  - Followed by chloride ion transport
  - Leads to osmotic swelling and bursting of vesicles
- PEI (poly ethylene Imine) is a widely used polymer for this phenomenon
- Conjugation of peptides that can create pores in endosomal membrane also been proposed
- However, the maximum reported efficacy of endosomal escape is <10%

Brock, D.J. et al. (2018) Traffic 19, 421–435

Proton Sponge effect



Membrane impermeable calcein and has low fluorescence at low pH



- Ensures that signal is presented is through MHCI and MHCII both

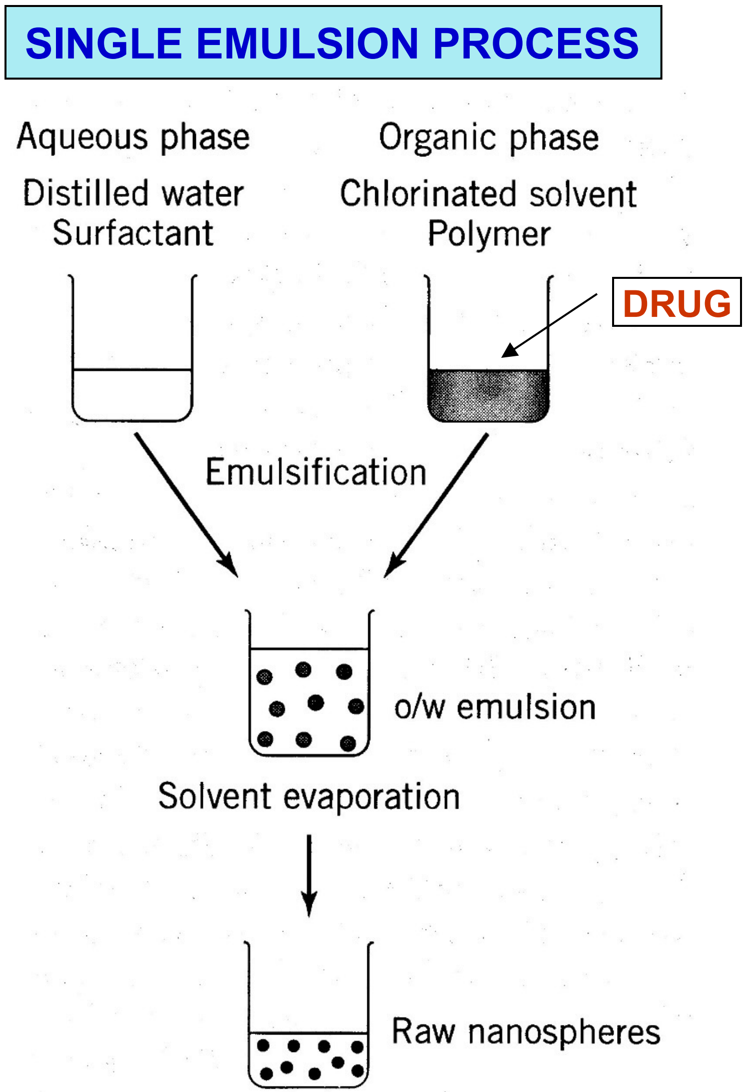
The Journal of Biological Chemistry 278, 44826-44831.

Macromol. Biosci. 17, 1600248

Particle Synthesis

- Chemical methods:
  - For polymer particles: Involves polymerization (interfacial or emulsion) of monomers during the synthesis of particles
  - For Metal particles: Reduction/oxidation/Crystallization from salts
- Physical methods:
  - Controlled “precipitation” of polymer **emulsions (single or double emulsions)**
  - Complex coacervation / coacervation
  - Solvent evaporation
  - Solvent removal (extraction)
  - Hot melt
  - Spray-drying
  - Phase separation
  - Salting out

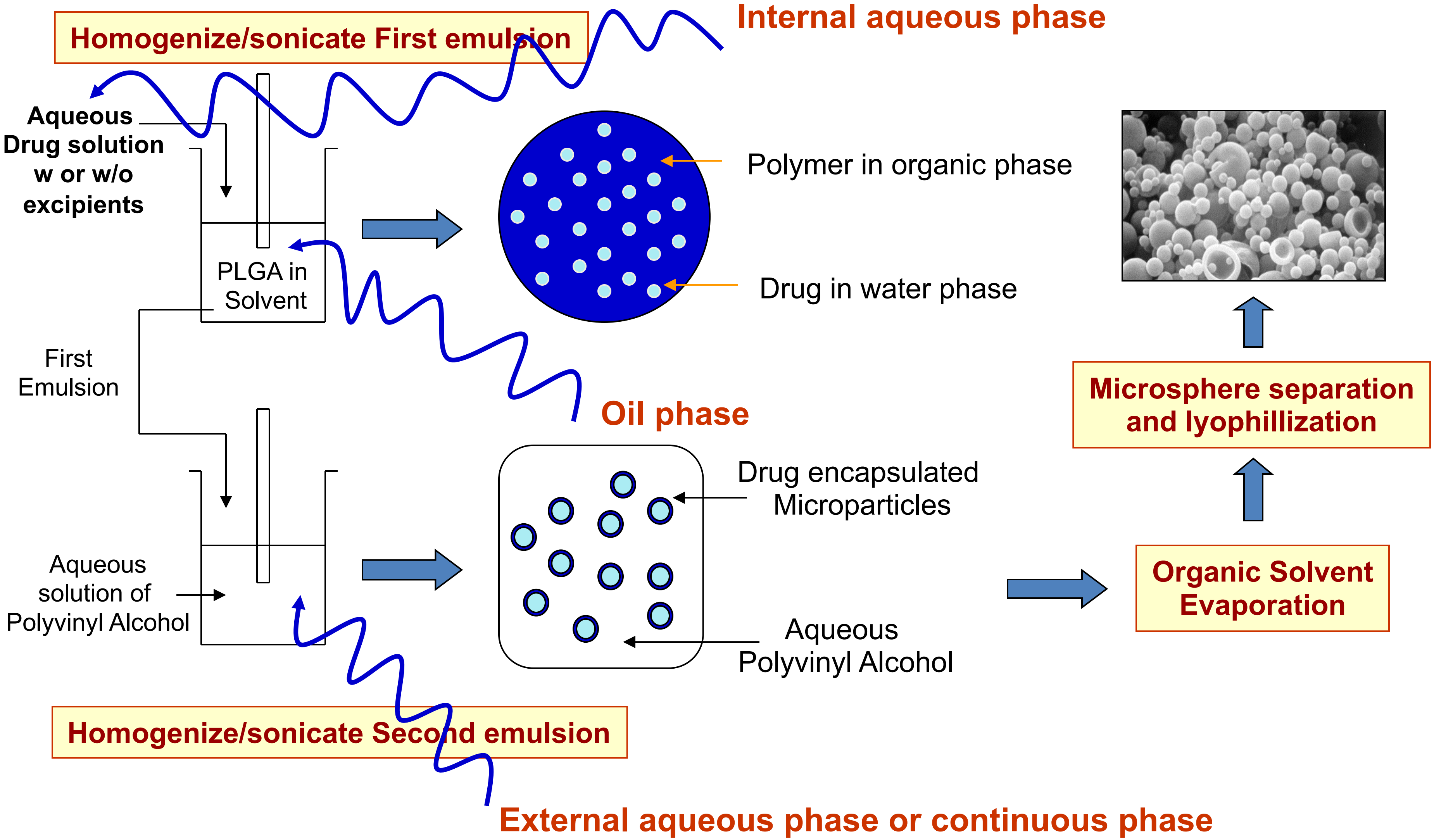
PARTICLE SYNTHESIS: SOLVENT EVAPORATION



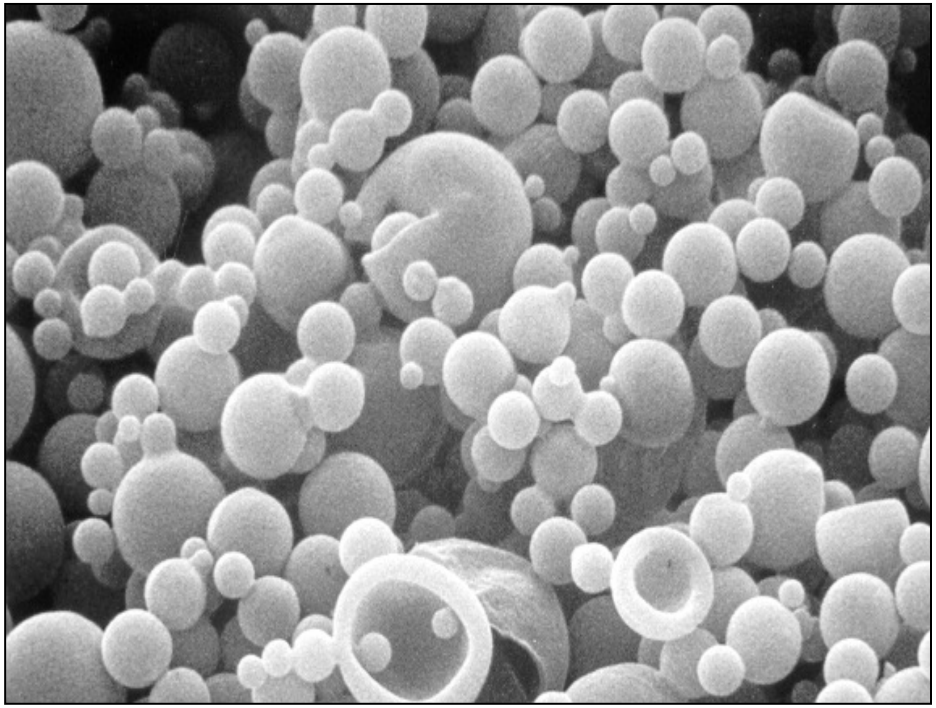
- PLGA/PLA microparticles as example
- Oil-in-water (o/w) emulsion (could also be oil-in-oil emulsion) → first oil phase must not be soluble or miscible in second oil phase
- Produces a solid matrix type particles (no capsules or hollow particles)
- Drug must be soluble or dispersible in organic solvent phase.
- Polymer dissolved in a volatile organic solvent → most commonly methylene chloride or dichloromethane (DCM). Other solvents used are chloroform, ethyl acetate etc.
- The polymer-drug solution/dispersion is emulsified in a large volume of aqueous phase → water containing surfactants, most commonly polyvinyl alcohol (PVA)
- Emulsion is stirred (could be under reduced pressure and elevated temperature) for several hours to evaporate organic solvent and "harden" the emulsion droplets → droplets solidifies into particles

PARTICLE SYNTHESIS: SOLVENT EVAPORATION

TYPICAL W/O/W DOUBLE EMULSION PROCESS



PARTICLE SYNTHESIS: SOLVENT EVAPORATION



Parameters affecting particles

- Polymer type and molecular weight
- Polymer concentration in the oil phase
- Type of drug and method of incorporation (solid, liquid, suspension, hydrophilic, hydrophobic etc.)
- Organic solvent used and polymer solubility
- Type and amount of surfactant in the external phase
- Internal aqueous phase / organic solvent ratio
- Rate of stirring and type of mechanical force – Increased stirring, decreased particle size
- Evaporation rate / temperature / pressure
- Extraction rate (volume, temperature, etc.)

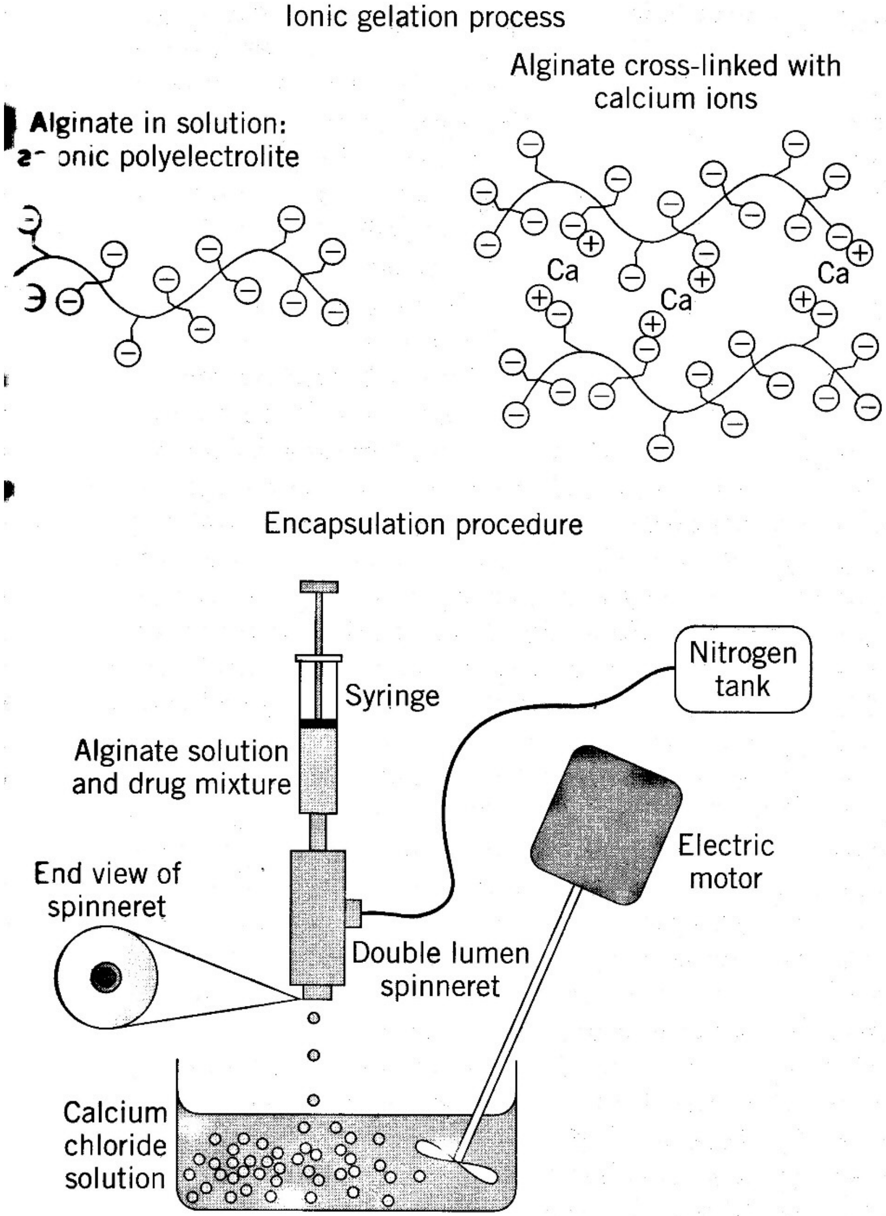
Single or double emulsion processes: Solvent extraction/removal method

- **Most solvents used to dissolve the polymer [Oil phase] has some solubility (generally low) in water.**
- Emulsion can be added to a very large volume of aqueous solution (with or without surfactant) → volume should be large enough for all the organic solvent to be "soluble" in the water phase
- S**olvent is rapidly extracted out of the polymer phase into the external continuous phase**
- Hardens the microparticles rapidly
- **Internal particle structures / porosities etc. can be altered by choosing evaporation versus extraction.**
- **Disadvantage: large volume required for solvents like DCM → difficult to scale up for industrial productions**
  - For example: **DCM is soluble In water up to ~1.5% w/w. So to "extract" 10 ml of DCM rapidly by this process, the volume of the external aqueous phase must be greater than 660 ml**

Emulsion based methods: Short comings

- High polydispersity
- Biomolecule denaturation
- Low encapsulation efficiency for water soluble drugs
- Difficult to synthesize sizes below 150nm

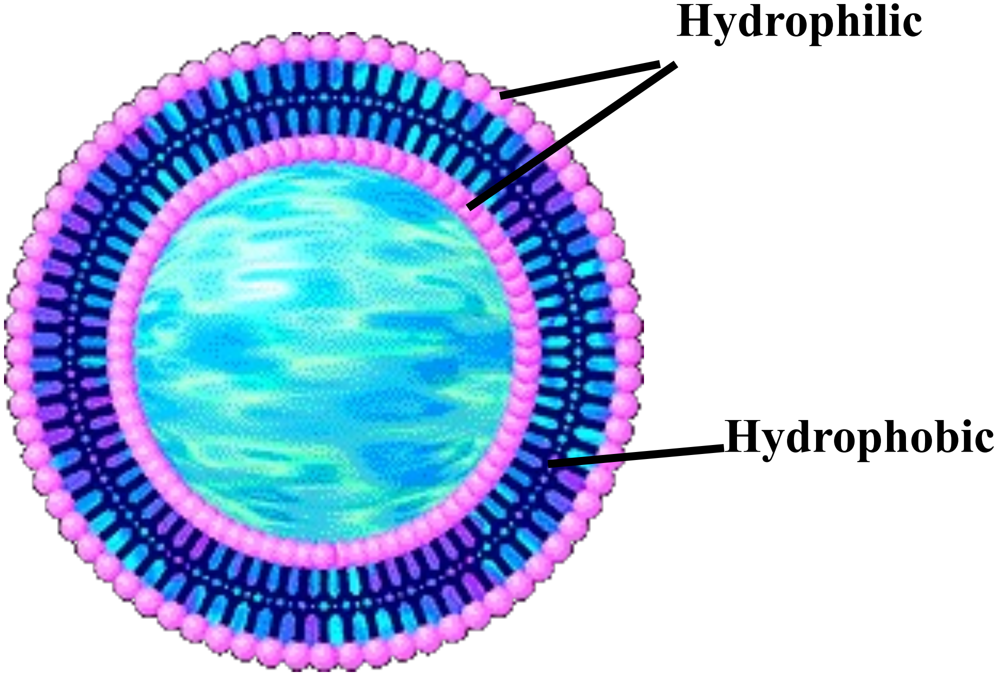
PARTICLE SYNTHESIS: GELATION



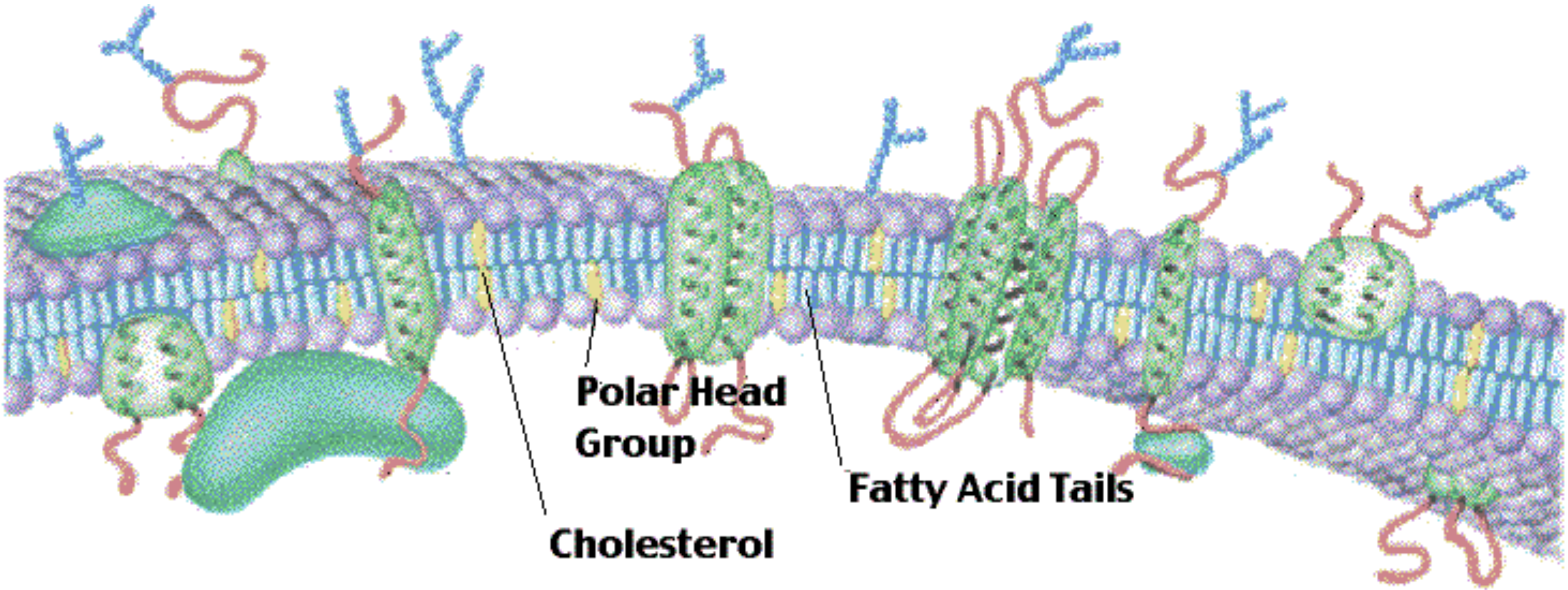
- A hydrogel microparticle: Physical ionic hydrogel composed of anionic polyelectrolyte (alginate) crosslinked with a bivalent cation (calcium)
- Drugs, cells, tissues (islets) etc. have been encapsulated within these microparticles
- Fairly mild conditions (except exposure to high calcium concentration could be detrimental to cells)

What is a liposome?

phospholipid bilayer

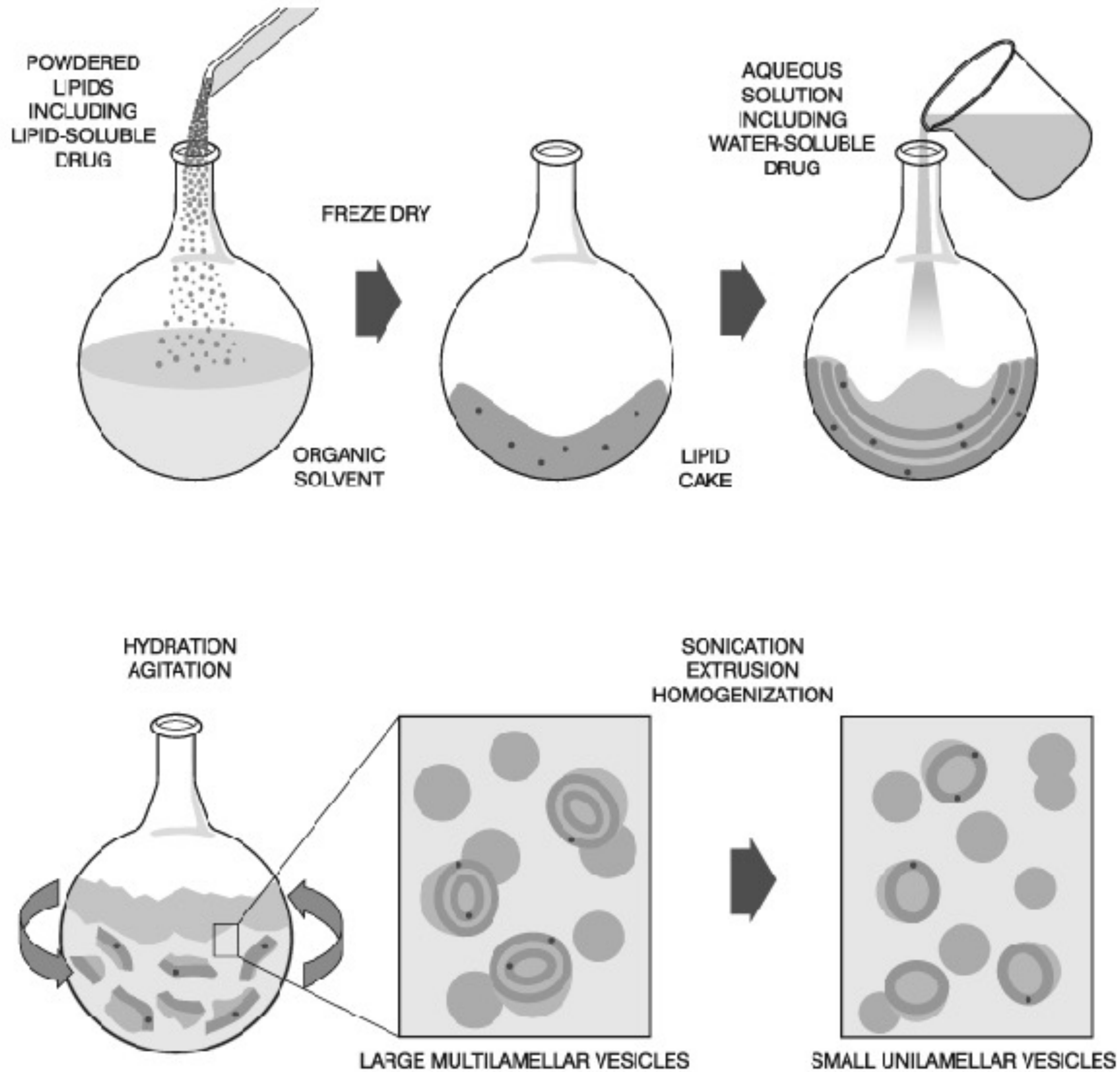


Cell Membrane Phospholipids



Phospholipids are the major structural components of biological membranes such as the cell membrane

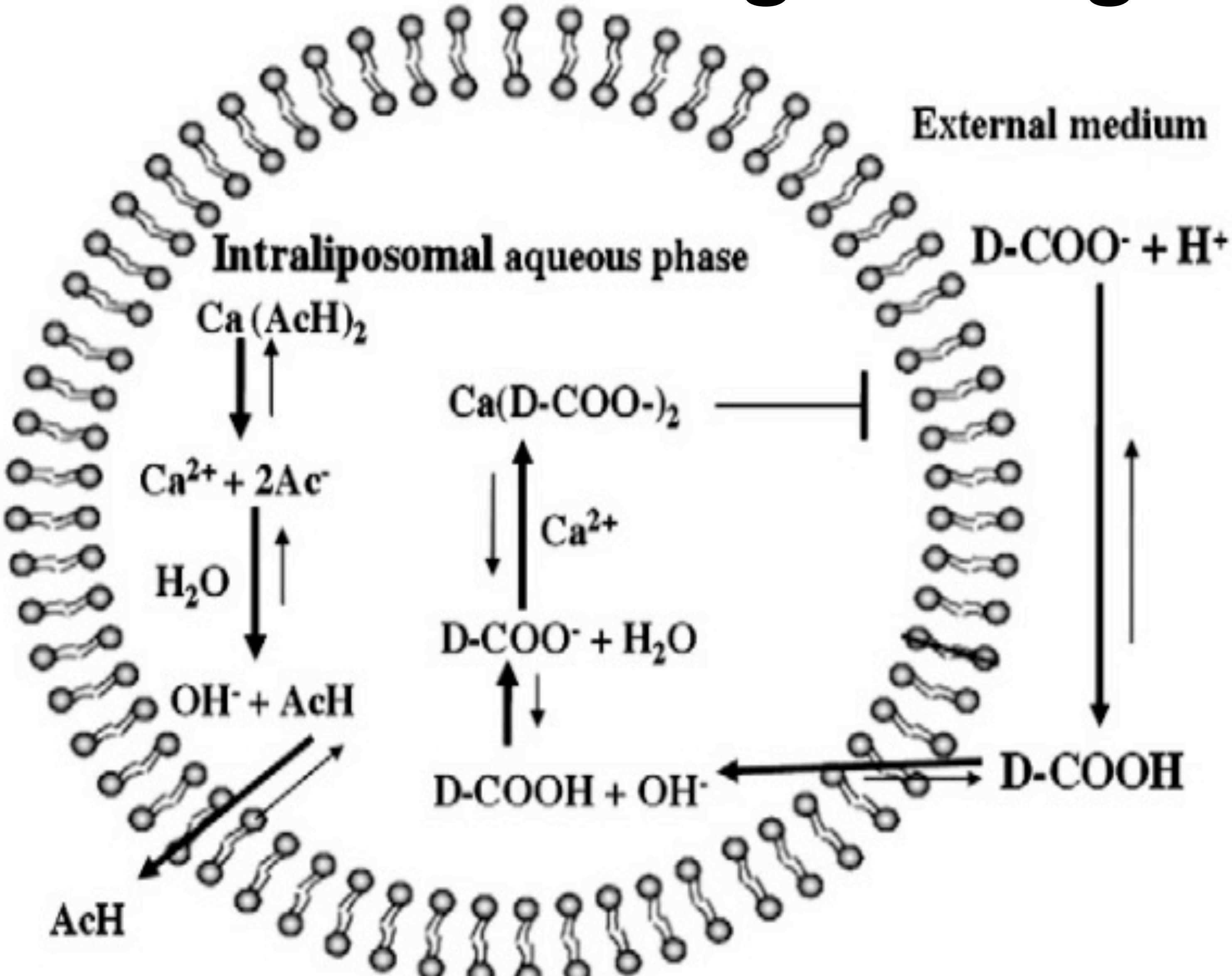
Liposome synthesis and drug encapsulation



Release from liposome

- Lipid membranes are fluidic at high temperature (transition temperature) and hence release the drug
- Lipids are chosen such that they are fluidic just above 37^{o}C
- Liposomes can also burst when in contact with large amphipathic compounds such as proteins

Remote loading of drug

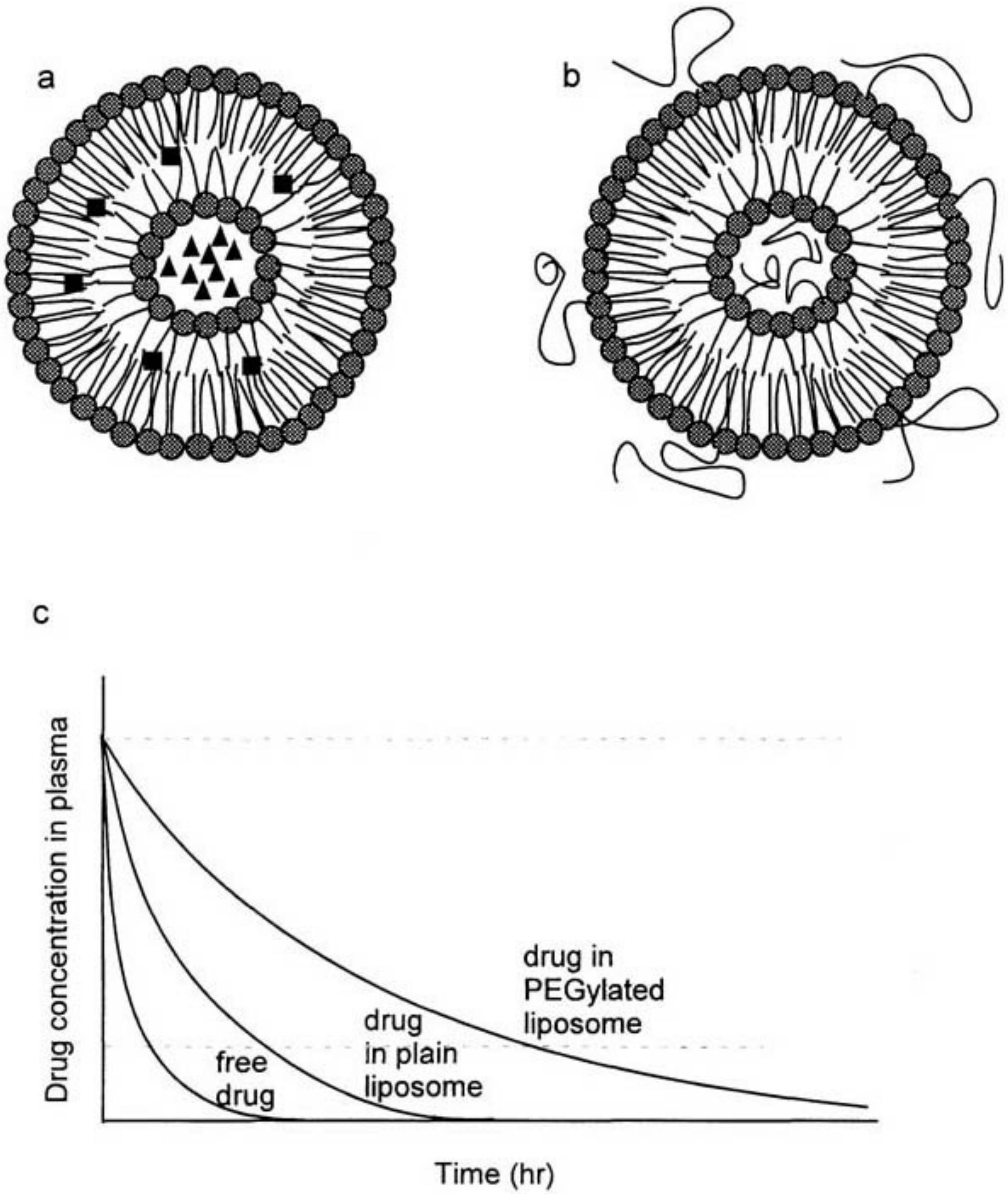


Journal of Controlled Release 139 (2009) 73–80

Remote loading of drug

- Loading is done on preformed liposomes
- Neutral drug will diffuse through the membrane and will be converted to charged species incapable of diffusing out
- Only amphipathic weak acids and bases fit these requirements
- pH and Ion gradients were created by salts of either weak bases (e.g., ammonium) or weak acids (e.g., acetate). These ions can be present (depending on pH) as charged and uncharged species. Such ions can traverse the liposome membrane only in their uncharged form.
- Hydrated in either ammonium sulfate (for bases' loading), or calcium acetate (for acids' loading)

Stealth Liposomes: Half lives



Drug Delivery: Engineering Principles for drug therapy by Mark Saltzman, 2001

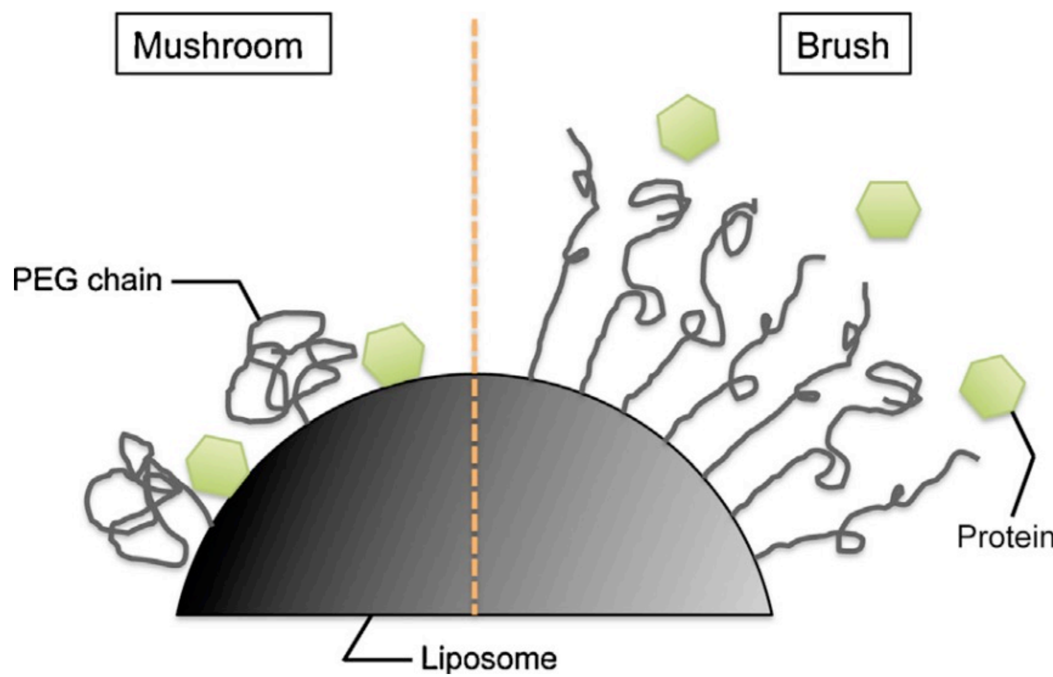
Stealth Liposomes: Half lives

Liposomes[b]		Half life (min)
3 mol%	PEG/PE	80
”	PEG2/PE	140
”	PAcM/PE	40
”	PVP/PE	90
7 mol%	PEG/PE	230
”	PEG2/PE	230
”	PAcM/PE	130
”	PEV/PE	170
Without any polymer		10

Half life of "stealth“ Liposomes in mice circulation after incorporation of linear PEG, branched PEG, PAcM and PVP

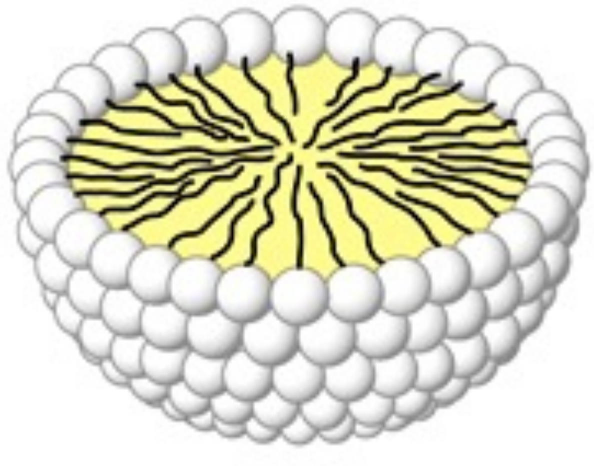
PEGylation: Brush vs Mushroom

- Brush type conformation is considered better
- Applicable for any type of particle/surface



Colloids and Surfaces B: Biointerfaces 124 (2014) 17-24

Micelles



- Aggregate of amphipathic molecules in water, with the nonpolar portions in the interior and the polar portions at the exterior surface, exposed to water.

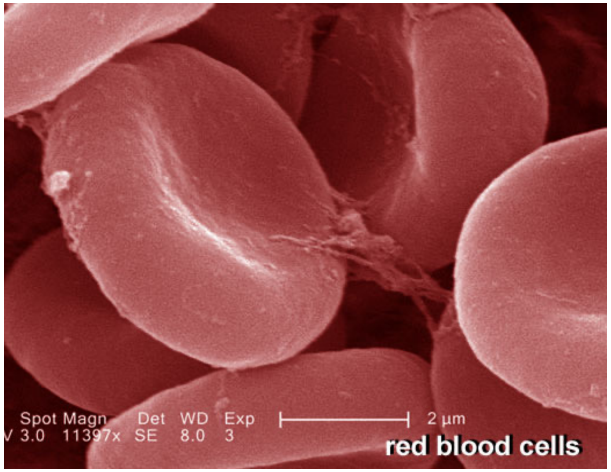
- Amphiphilic molecules form micelle above a particular concentration which is called as critical micellar concentration (CMC).
- Hydrophobic drugs can be encapsulated/solubilized, into inner core
- Micelle formation is favored when the cross-sectional area of the head group is greater than the acyl side chains

api.ning.com/files/.../NanotechnologyBasedDrugDelivery.ppt

Effect of particle shape

- Natural occurrence of diversely shaped entities like virus, bacteria, red blood cells
- Non-spherical shapes: large contact area to adhere to surfaces
- Rod: allows fast motility in bacteria
- Disc shaped RBCs: marginate in blood vessels

Bacteria



Red Blood Cells

http://www.bacteriapictures.net

http://www.sfdm.scad.edu

Young *et al.* Curr Opin Microbiol 10 (2007)

Making polymeric micro- and nanoparticles of complex shapes

Julie A. Champion, Yogesh K. Katare, and Samir Mitragotri*

Department of Chemical Engineering, University of California, Santa Barbara, CA 93106

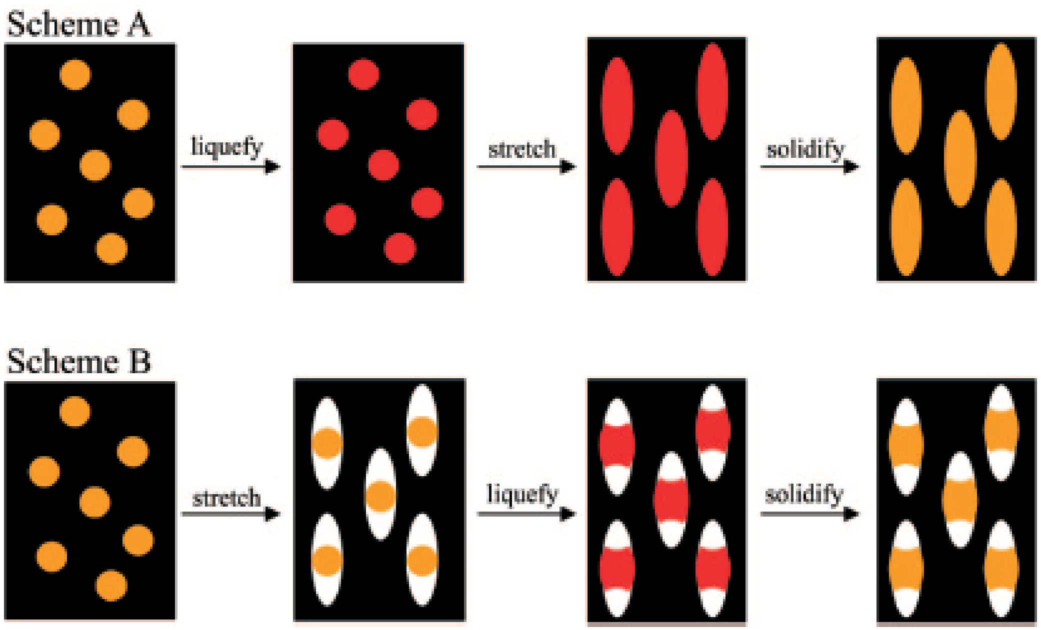


Fig. 1. Methods used for making particles with different shapes can be categorized into two general schemes. (*Upper*) Scheme A involves liquefaction of particles by using heat or toluene, stretching the film in one or two dimensions and solidifying the particles by extracting toluene or cooling. The example shown here produces elliptical disks. (*Lower*) Scheme B involves stretching the film in air to create voids around the particle, followed by liquefaction using heat or toluene and solidification. The example shown here produces barrels.

PNAS | July 17, 2007 | vol. 104 | no. 29 | 11901–11904

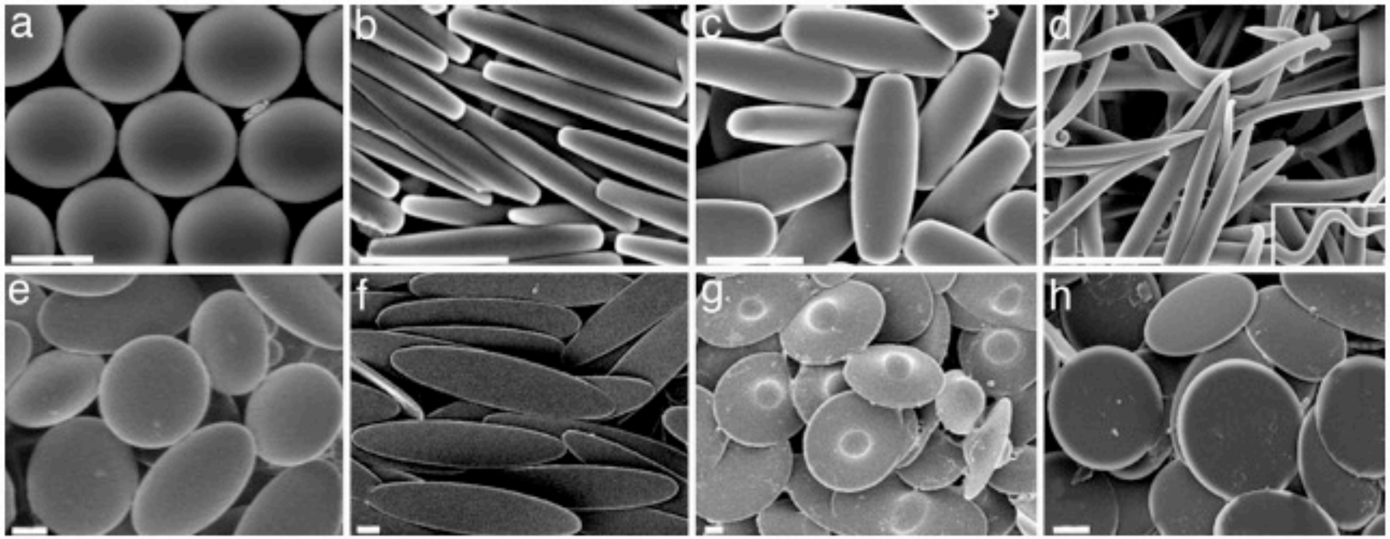
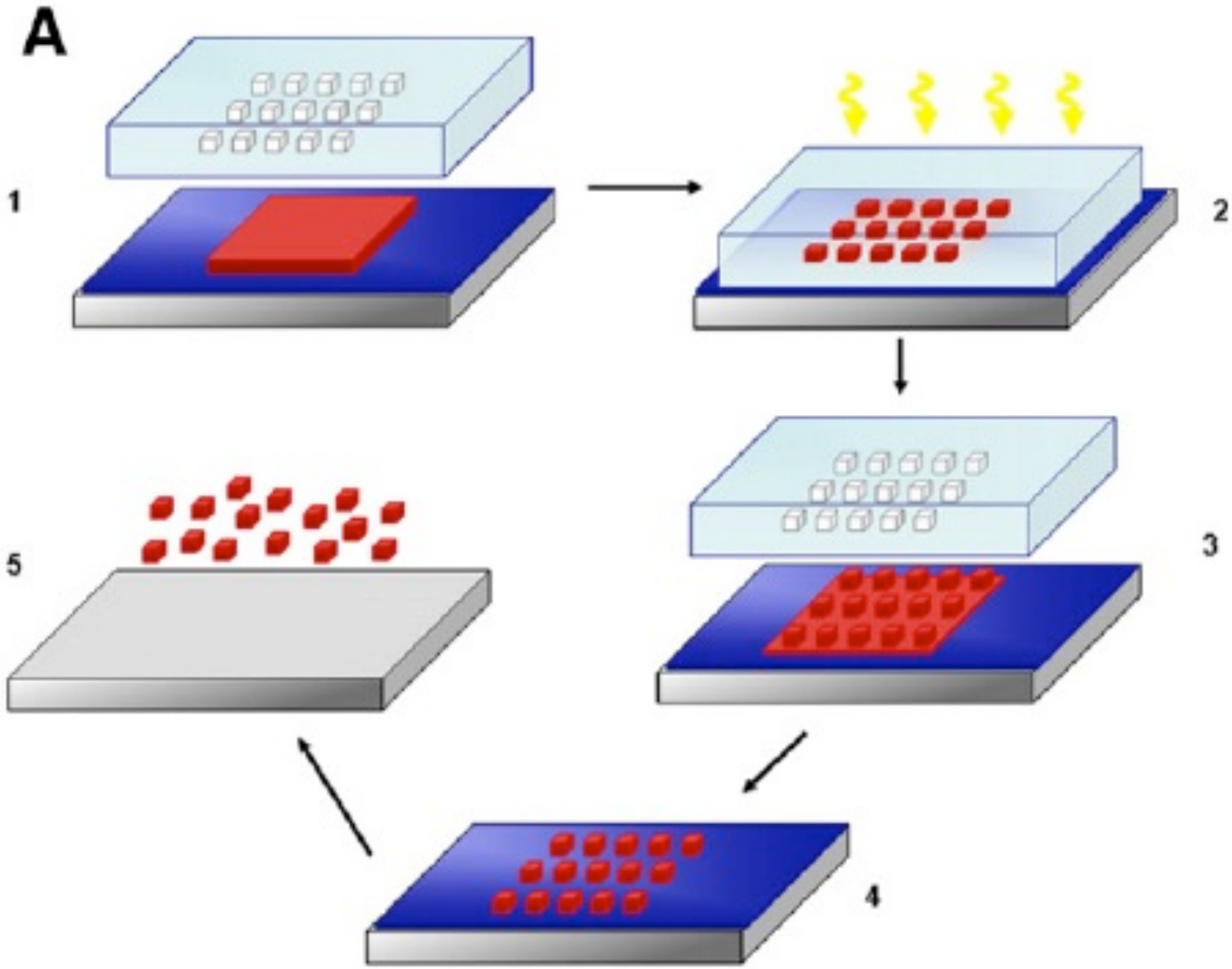


Fig. 2. Micrographs of shapes made by using scheme A. (*a*) Spheres. (*b*) Rectangular disks. (*c*) Rods. (*d*) Worms. (*e*) Oblate ellipses. (*f*) Elliptical disks. (*g*) UFOs. (*h*) Circular disks. (Scale bars: 2 μm.)

Journal of Controlled Release 125 (2008) 263 – 272

Nanoimprint lithography based fabrication of shape-specific, enzymatically-triggered smart nanoparticles

Luz Cristal Glangchai [a,1], Mary Caldorera-Moore [a,1], Li Shi [b,c,d,*], Krishnendu Roy [a,c,d,*]



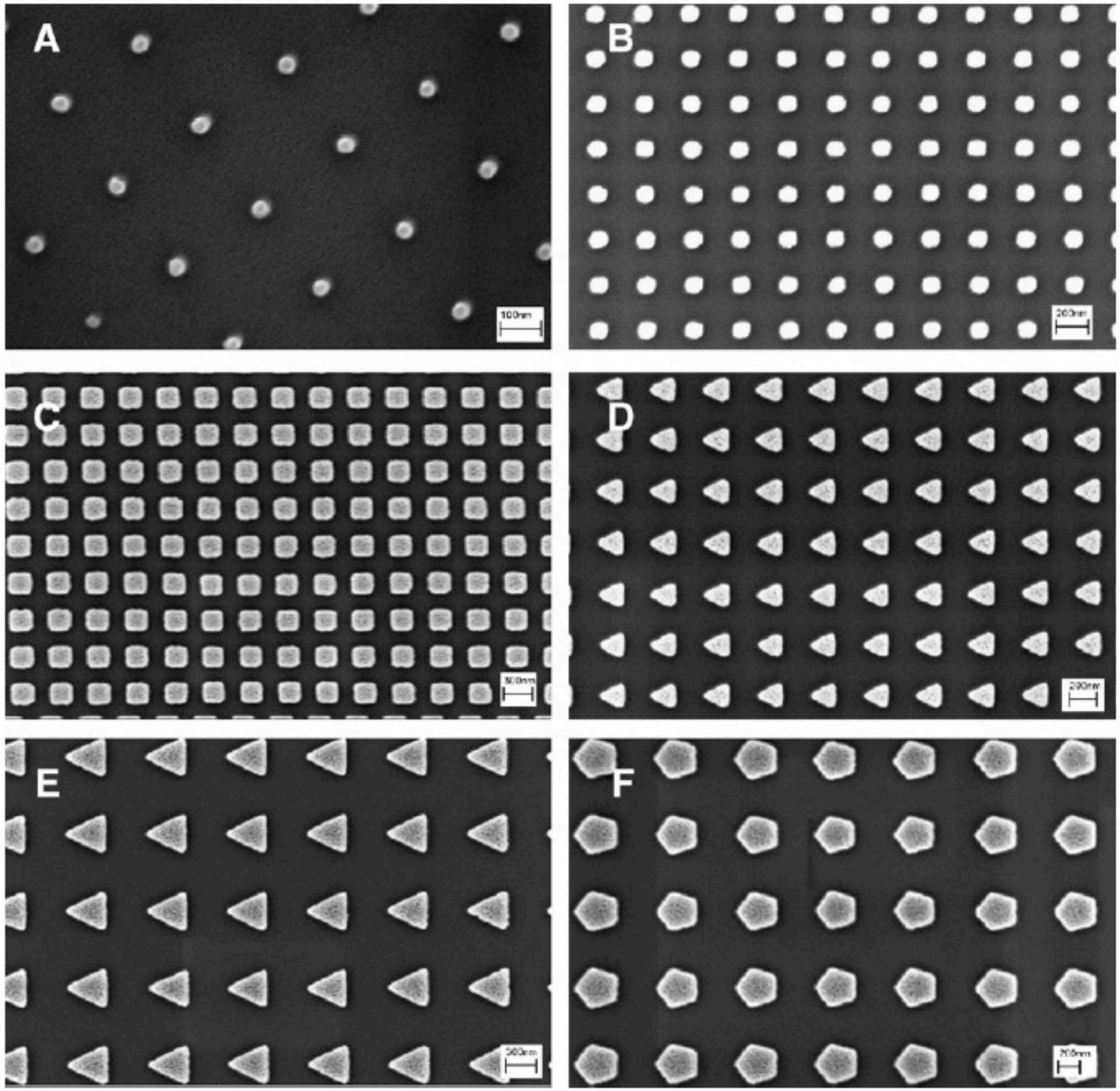
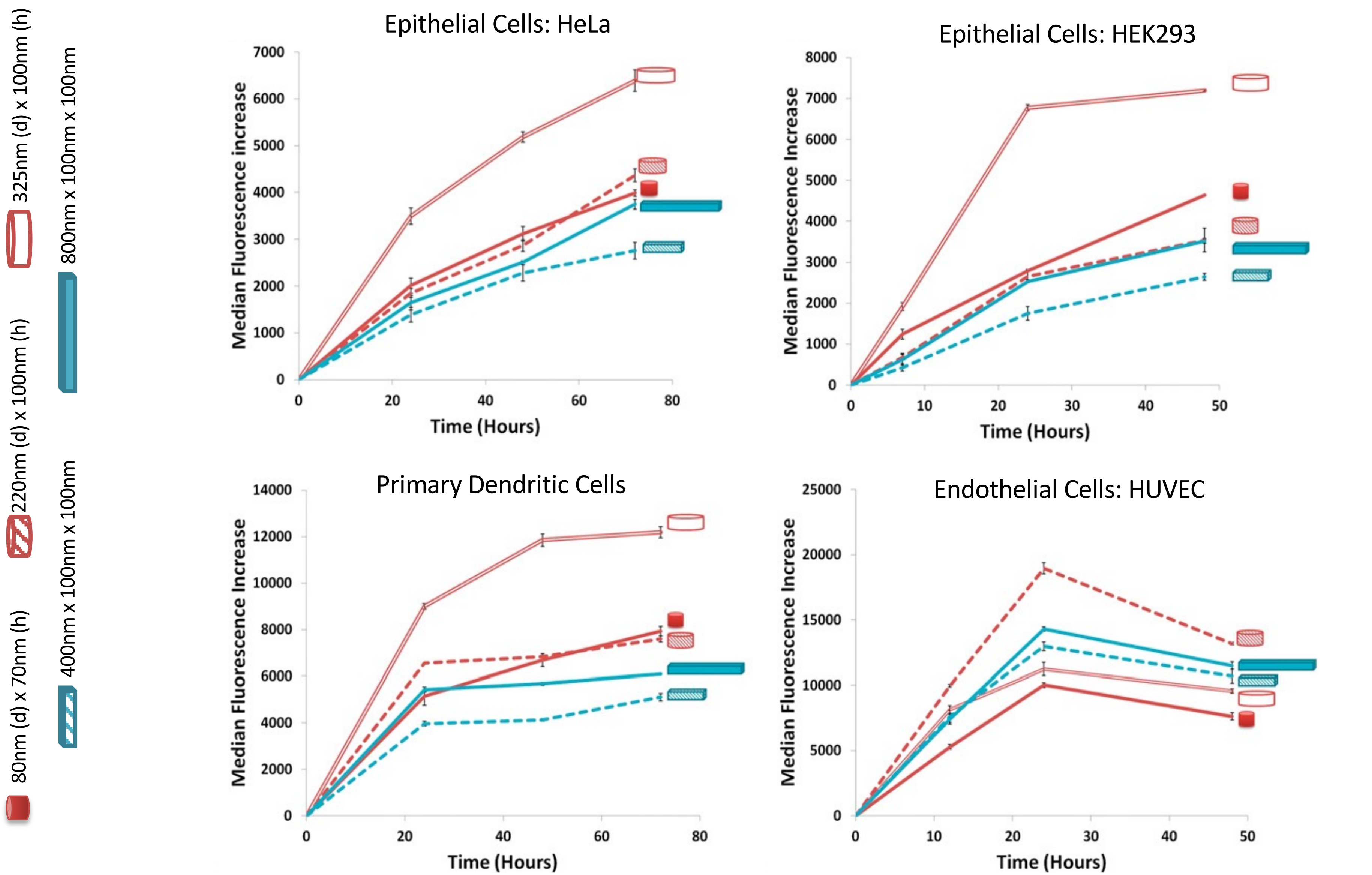


Fig. 3. SEM images of S-FIL imprinted (100% w/v, MW 3400) PEGDA nanoparticles: (A) 50 nm squares (scale bar=100 nm), (B) 100 nm squares (scale bar=200 nm), (C) 200 nm squares (scale bar=300 nm), (D) 200 nm triangles (scale bar=200 nm), (E) 400 nm triangles (scale bar=300 nm), and (F) 400 nm pentagonal particles (scale bar=200 nm).

Mammalian cells uptake high aspect ratio discs



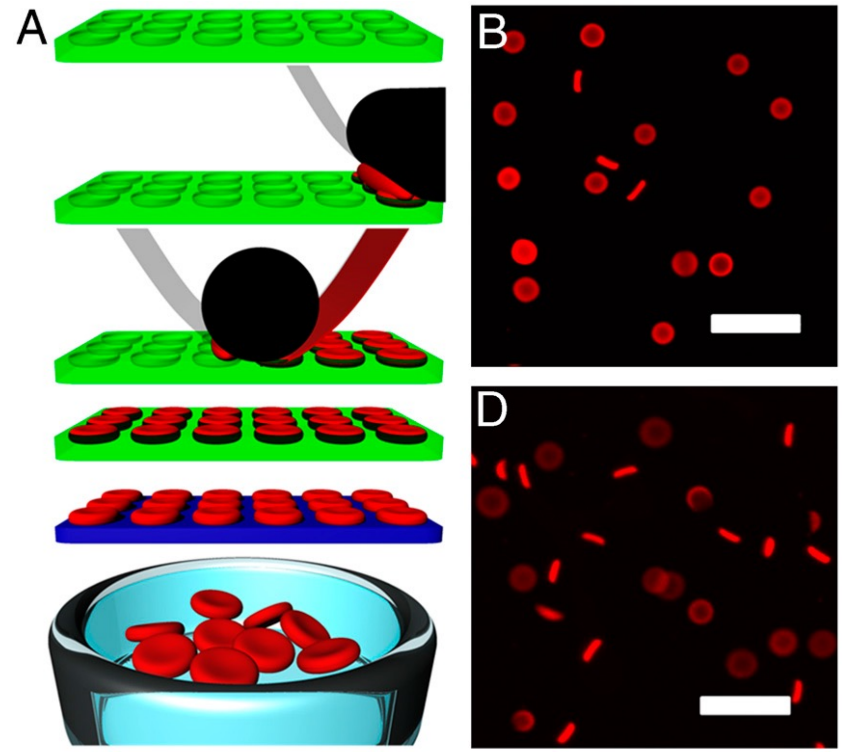
Agarwal *et al. PNAS* 110 (2013)

Charge of particle

- In body, particles encounter all kinds of charge
  - Cell membrane are slightly negatively charged
  - Serum proteins have all charges but predominantly anionic
- For delivery to mammalian cells in-vitro, positive charged particles are considered better
- In-vivo, positively charged particles are rapidly adsorbed by serum proteins and cleared
- Positively charged particles can also cause toxicity in the body
- Neutral and slightly negatively charged particles are able to circulate in the blood for long time
- However, the understanding is still not complete and certain applications may require use of positively charged particles

Elasticity of particle

- Elasticity has also been reported to have profound effect in circulation times in blood
- The best known example is RBCs, which are known to circulate in blood for months
- For low modulus, larger particles (6.4 µm) were able to circulate in blood longer compared with their smaller counterparts (780 nm)

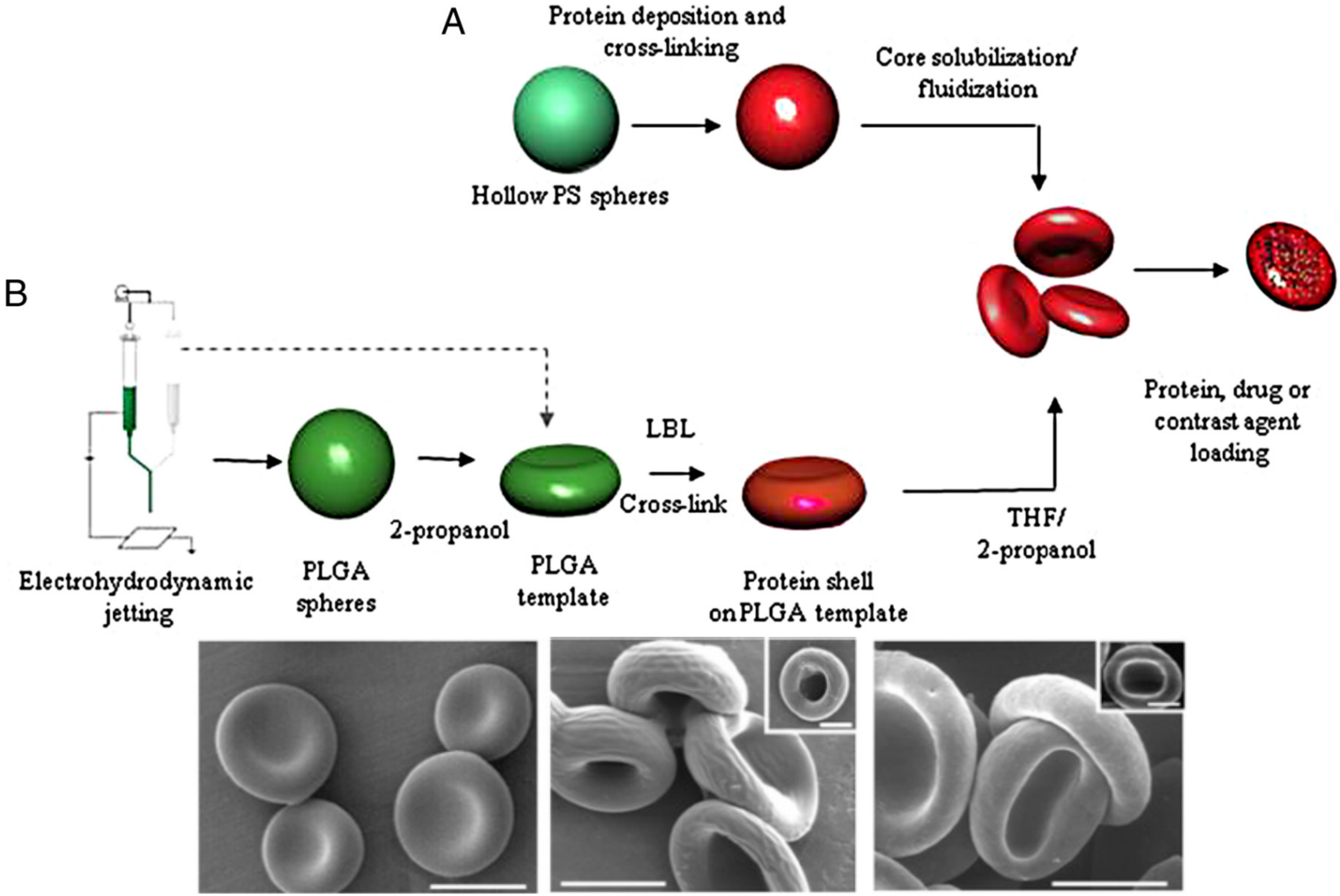


586–591 | PNAS | January 11, 2011 | vol. 108 | no. 2

Table 1. Modulus values and compartmental analysis of RBCM particles from intravital microscopy experiments

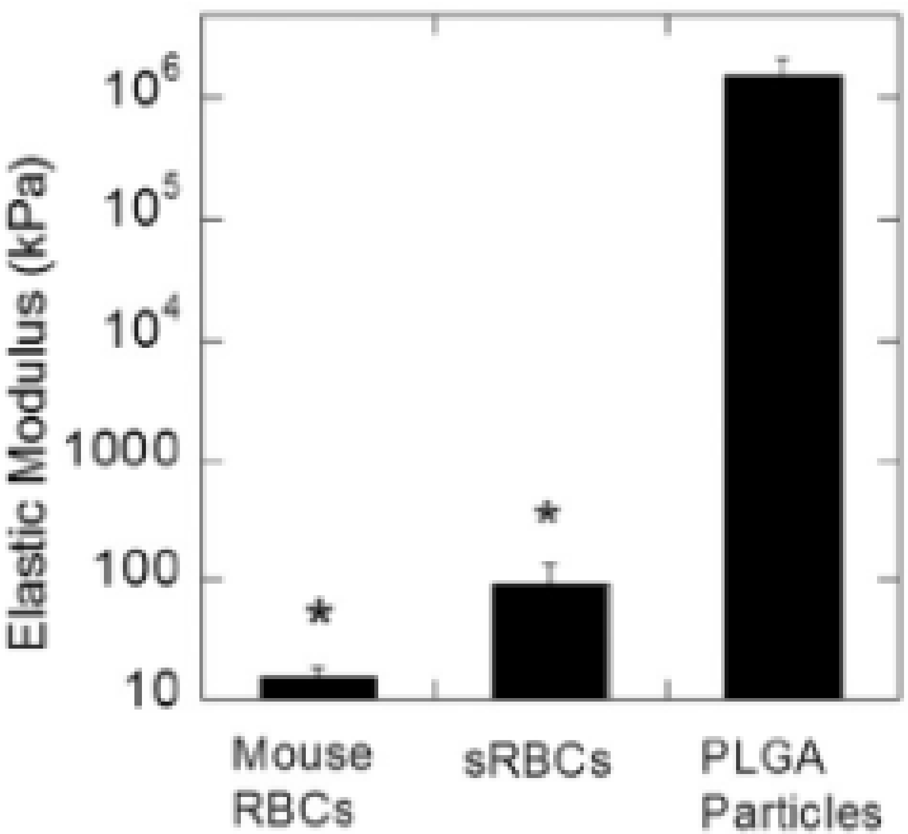
% Cross-linker	Modulus of bulk material, kPa	Distribution half-life, h	Elimination half-life, h	AUC, fluoresence* h	Average R^2
10%	63.9 ± 15.7	0.038 ± 0.0012	2.88 ± 0.92	0.65 ± 0.14	0.8966
5%	39.6 ± 10.4	0.066 ± 0.036	5.12 ± 2.17	0.76 ± 0.57	0.9029
2%	16.9 ± 1.7	0.15 ± 0.025	7.12 ± 0.82	1.35 ± 0.26	0.9468
1%	7.8 ± 1.0	0.35 ± 0.13	93.29 ± 31.09	15.44 ± 15.63	0.9330

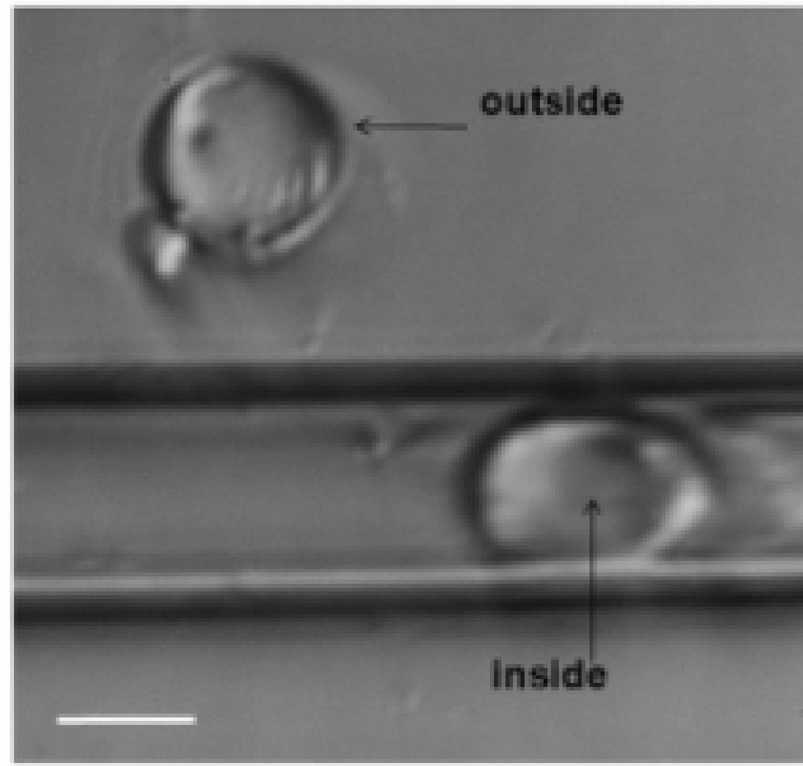
Elasticity of particle



electrohydrodynamic jetting   PLGA template particles   Cross-linked mouse RBCs

Elasticity of particle

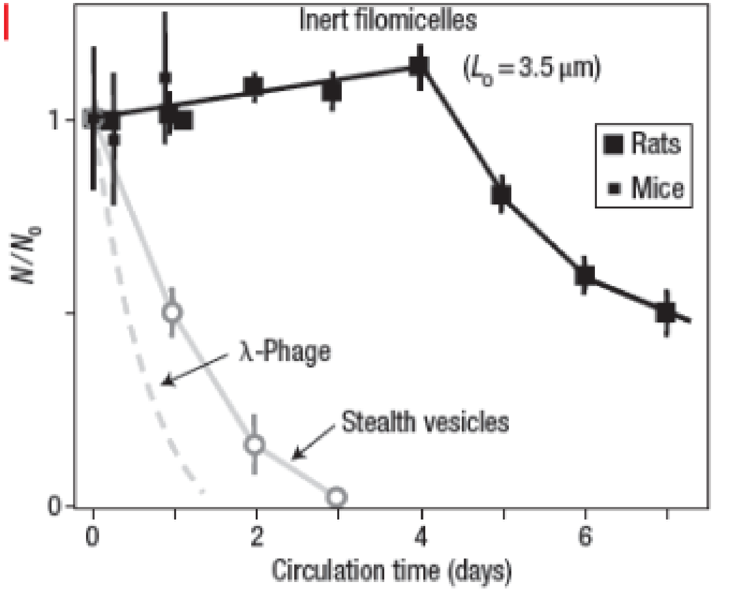




PNAS **December 22, 2009** vol. 106 no. 51 **21497**

Elasticity of particle

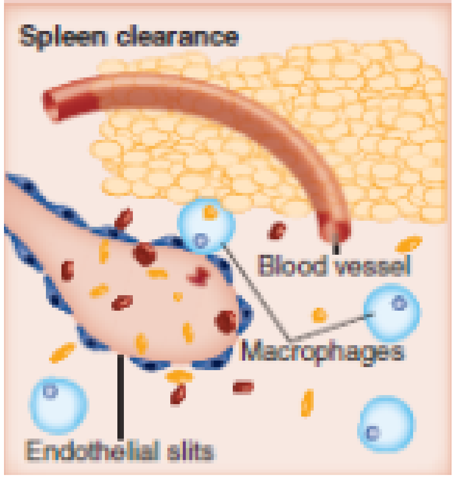
20-30 diameter, 5-8 um length



Crosslinked and rigid structures of similar dimensions clear rapidly

Reason?

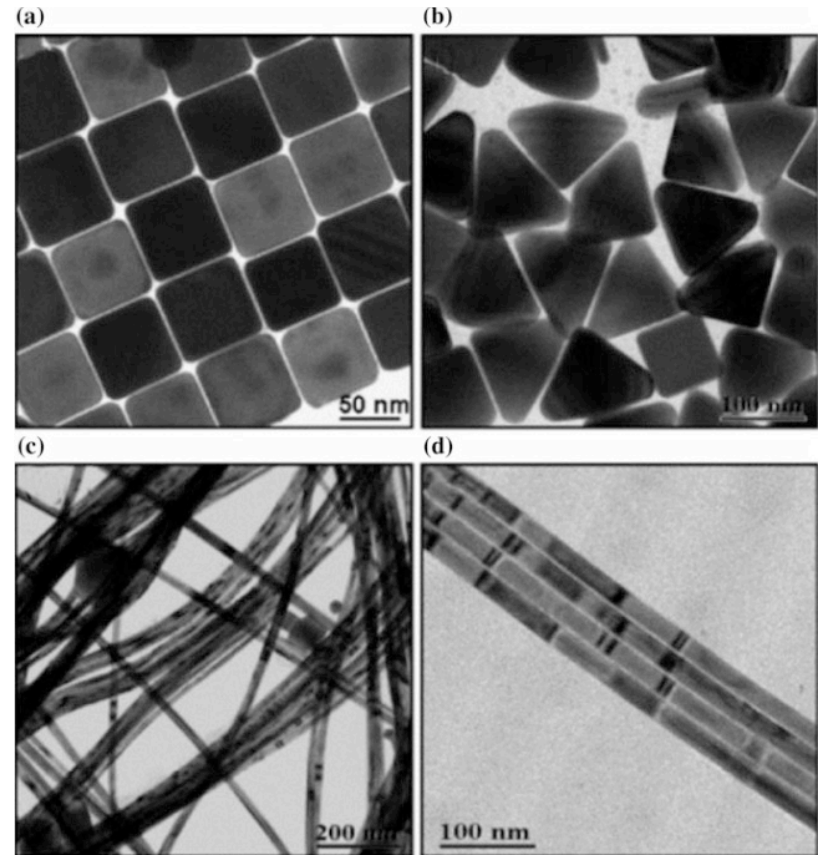
Particles are able to squeeze through the fenestrations in spleen and avoid uptake by immune cells

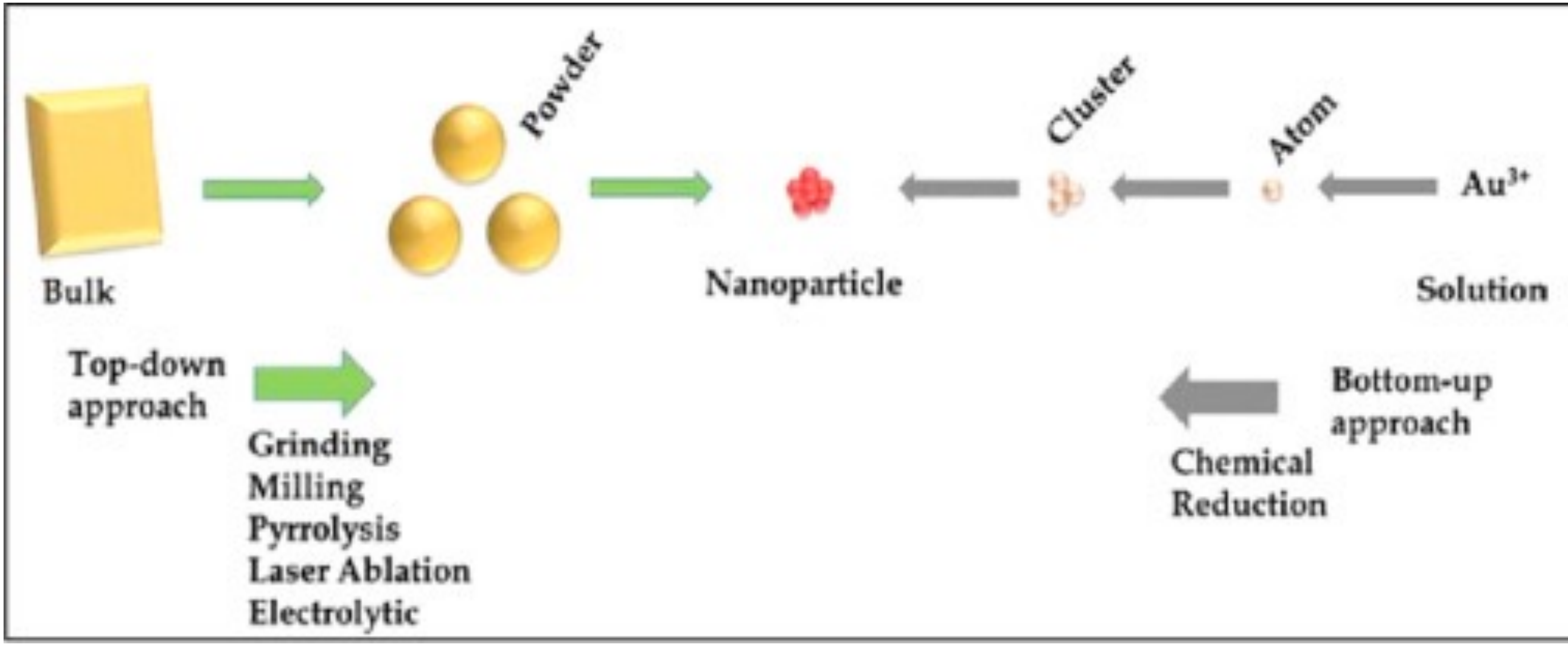


Nat. Nanotechnol. 2(4), 249–255 (2007)

Metal particles

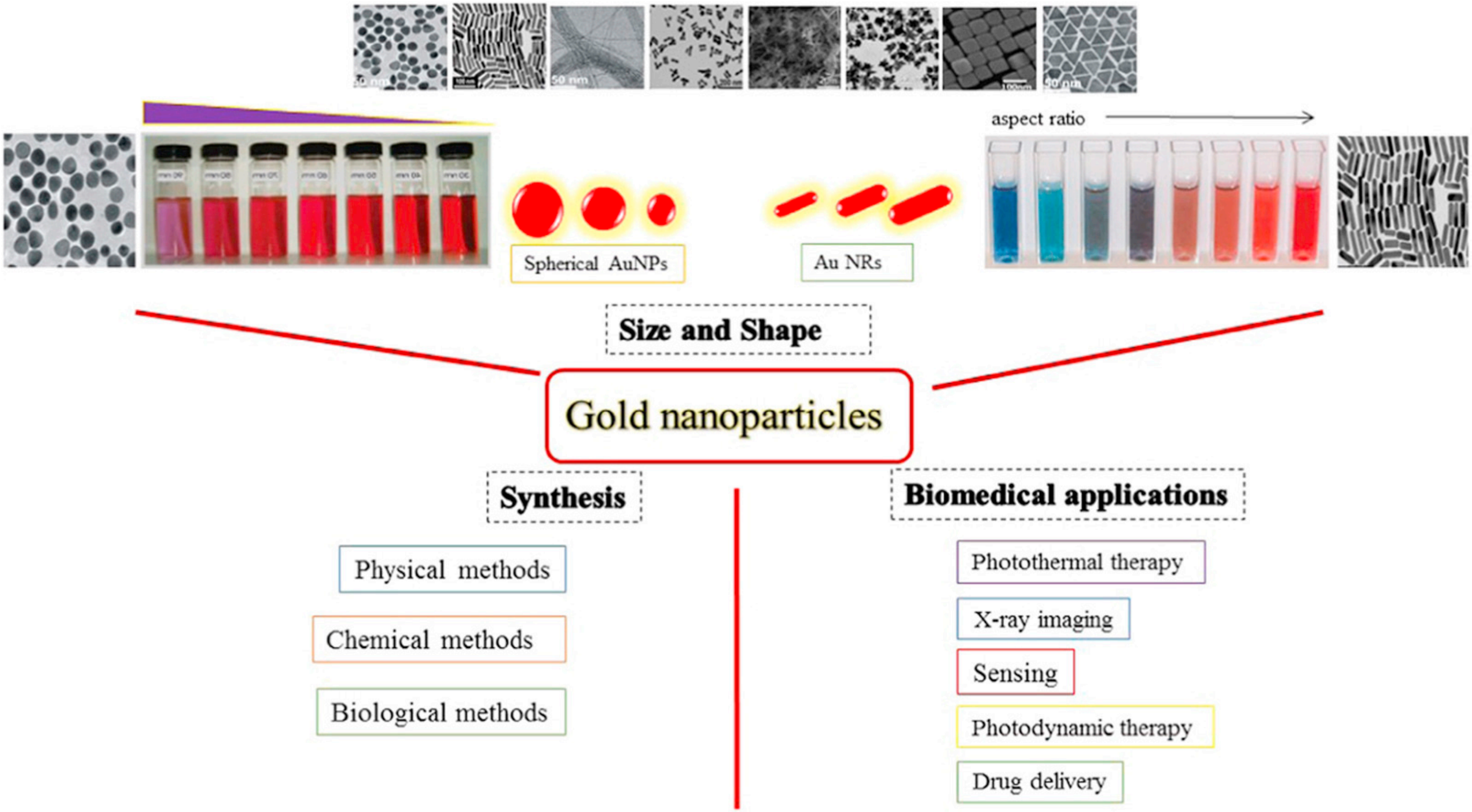
- Same concepts apply for metal particles
- Silver, gold, iron-oxide, quantum dots
- Widely used for imaging application
- Optical response is tunable based on size and shape of the particles: Heating, fluorescence
- Drug can also be coated/conjugated on the particle surface
- Non-degradable





- Synthesis is extremely monodisperse
- Well established techniques to synthesize various shapes
- Top down approaches include grinding, milling etc
- Bottom up approaches typically use chemical reduction

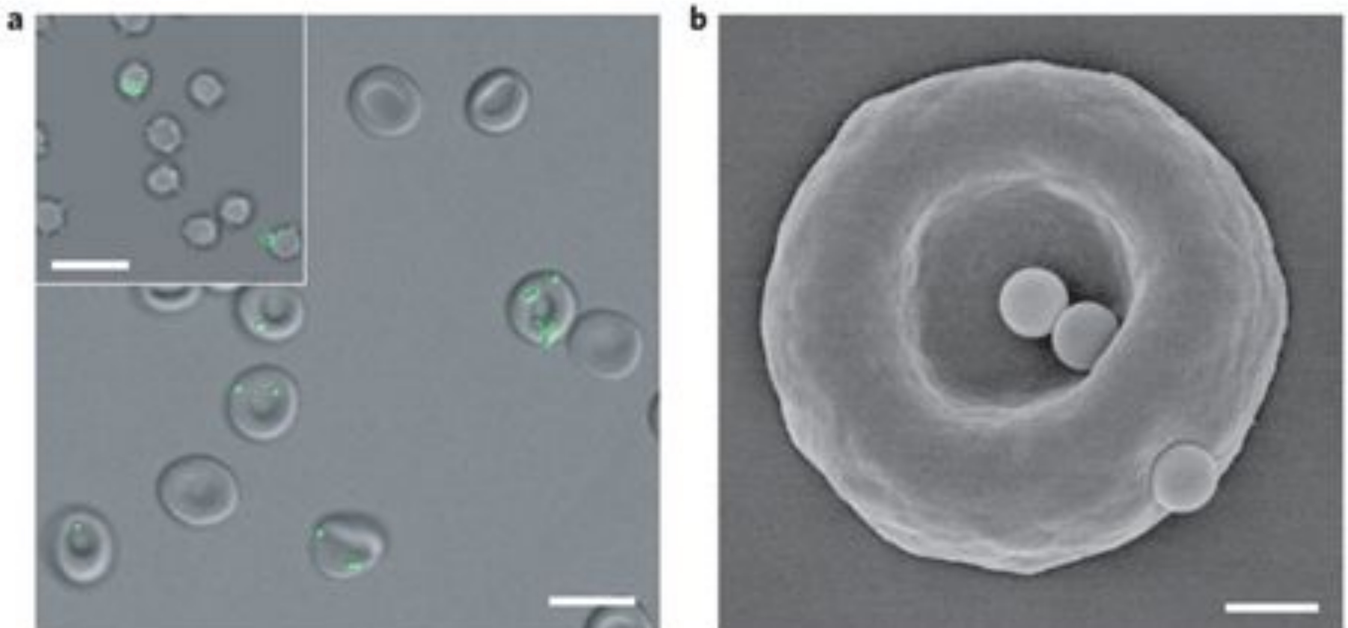
https://doi.org/10.1016/j.talanta.2018.02.088



https://doi.org/10.1016/j.talanta.2018.02.088

Particle hitch-hiking

- Particles can be conjugated to a target cell that prevents its clearance
- “Imparts” biological homing activity to particles
- RBCs and immune cells are the attractive targets
  - Adsorption
  - Chemical conjugation
- Need to be careful not to affect the host cell function



Nature Nanotechnology volume12, 589–594 (2017)